2020 MathWorks 中国汽车年会

务实的数字化转型
MathWorks和MATLAB如何帮助客户解决问题

周拥华
MathWorks中国区技术经理

MathWorks®

门铃的数字化转型

Digital technology

- HD video
- Motion detection
- Smartphone interface
- AWS Cloud

Amazon Acquires Ring, Maker of Video Doorbells

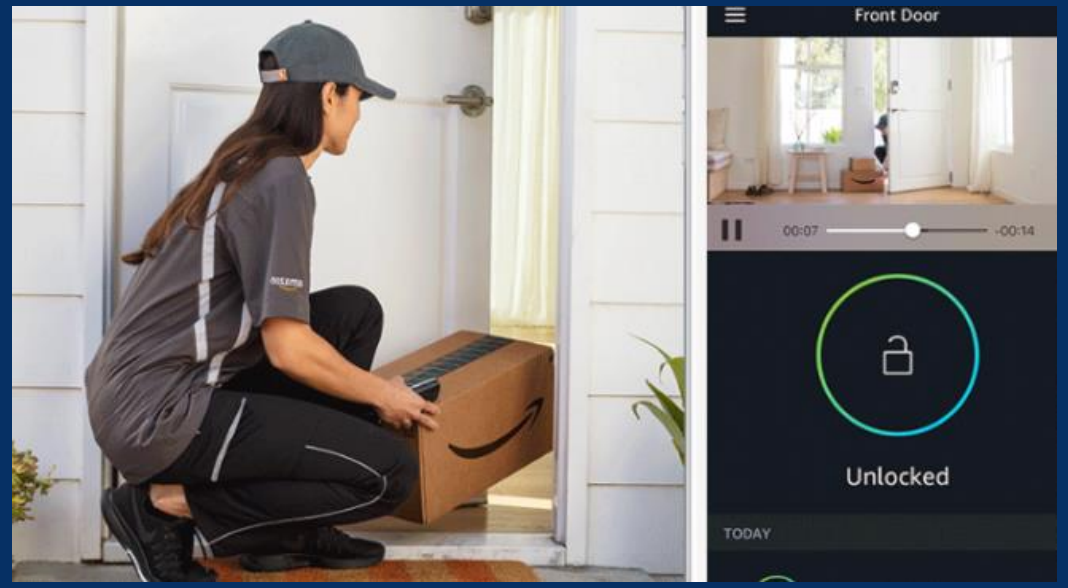
Front-door monitoring device plays to buyer's ambitions in home-security business

Business value

- Amazon buys Ring for $1.2 billion+ in 2018

New revenue opportunities

- "Ring Protect" subscription plans ($99-$499)
- Additional security with Ring Alarm kit
- More secure delivery through Amazon Key



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数字化转型的初衷

Do things better
Optimization

- Optimize design performance in-operation
- Predict when system needs maintenance
- Manage a fleet of connected systems

Do new things
Transformation

- Go into new industries and markets
- Expand into an entire platform service
- Provide unique value to your customer

The doorbell illustrates both types

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Plan and Pilot **Launch!**

Expected project duration

Actual project duration

Plan | Plan Some More | Pilot | Keep Piloting | **Launch?**

< 20% of organizations are on target with their digital transformation objectives

Source: McKinsey, *Can IT Rise to the Digital Challenge?*, October 2018.

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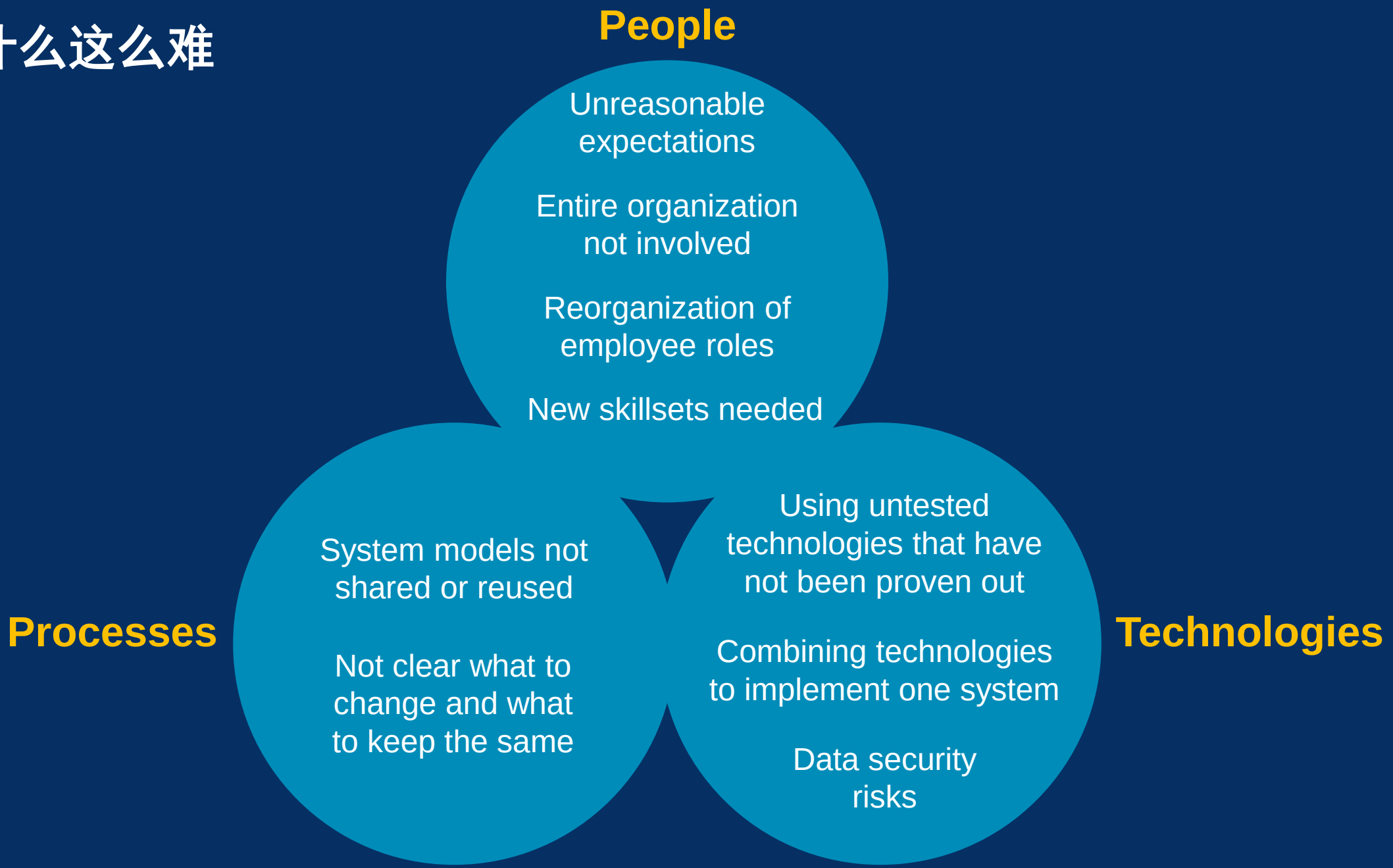
为什么这么难



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大家尝试过哪些解决方案

Big Bang Approach

Build complete infrastructure first

Value not delivered to customer

Risky

Pragmatic Approach

Build on models you already have

Extend beyond siloed use of data

Unleash untapped value

Siloed Approach

Each group works in own silo

Stuck in business model

Obsolete

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Pragmatic Digital Transformation

Systematic use of data and models
to **create** and **deliver** superior value to customers
throughout the entire lifecycle

数据中心化使工程工作变得更难

Field data

System data

User data

Environment data

Big Data

databricks hadoop CLOUDERA

Cloud Platforms

docker amazon web services Azure

Data diversity complexity

- Engineering, Scientific, and Field
- Business & transactional
- Noisy, Outliers, Missing data
- Time series synchronizing

Modern data management multiplies complexity

- Proliferation of data systems
- More siloes
- Cloud, on-premise, hybrid
- Big Data

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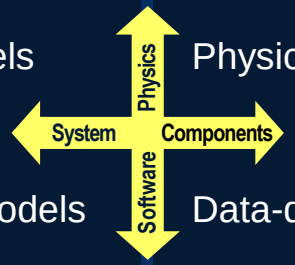
将模型拓展到系统生命周期

Closed-loop back to Development

System Requirements	System Functionality and Architecture	Subsystem Design	Subsystem Implementation	System Integration and Qualification	Operations and Sustainment
Use Cases Docs & models	Behavior models Architecture models	Physics-based Data-driven	C,C++ VHDL, Verilog GPU code Structured text	Model-based V&V Code-based V&V Certification workflows	Predictive maintenance Digital Twins



Digital Thread

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How MathWorks & MATLAB Help Customer Solving

围绕数据与模型进行开发的工作流程

Data Preparation

- Data cleansing and preparation
- Human insight
- Simulation-generated data

Modeling

- Model design and tuning
- Hardware accelerated training
- Interoperability

Simulation & Test

- Integration with complex systems
- System simulation
- System verification and validation

Deployment

- Embedded devices
- Enterprise systems
- Edge, cloud, desktop

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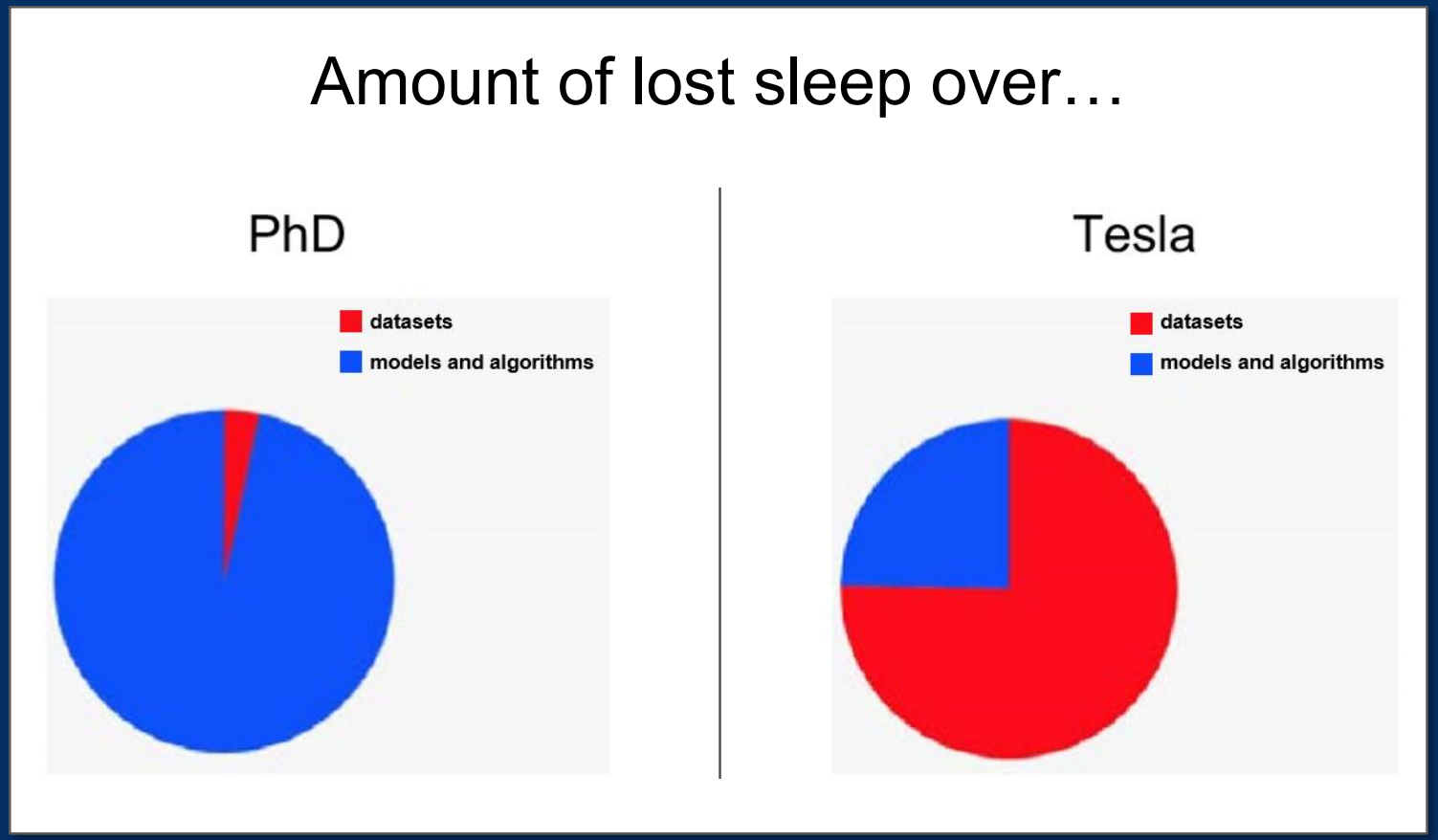
Data preparation represents most of your effort

Data Preparation

- Data cleansing and preparation
- Human insight
- Simulation-generated data



Source: Andrej Karpathy slide from TrainAI 2018

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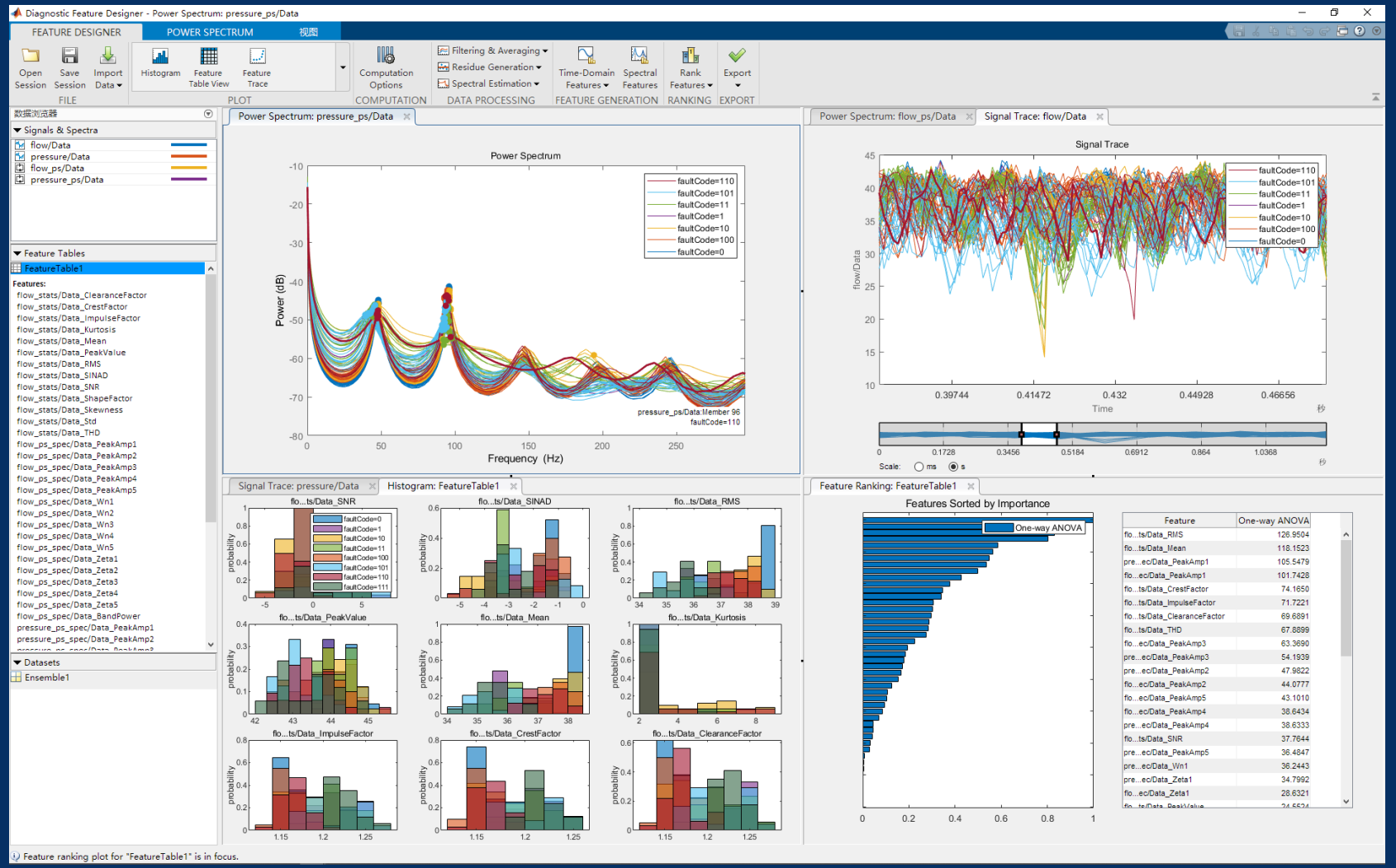
Automated Apps save you weeks to months

Data Preparation

- Data cleansing and preparation
- Human insight
- Simulation-generated data



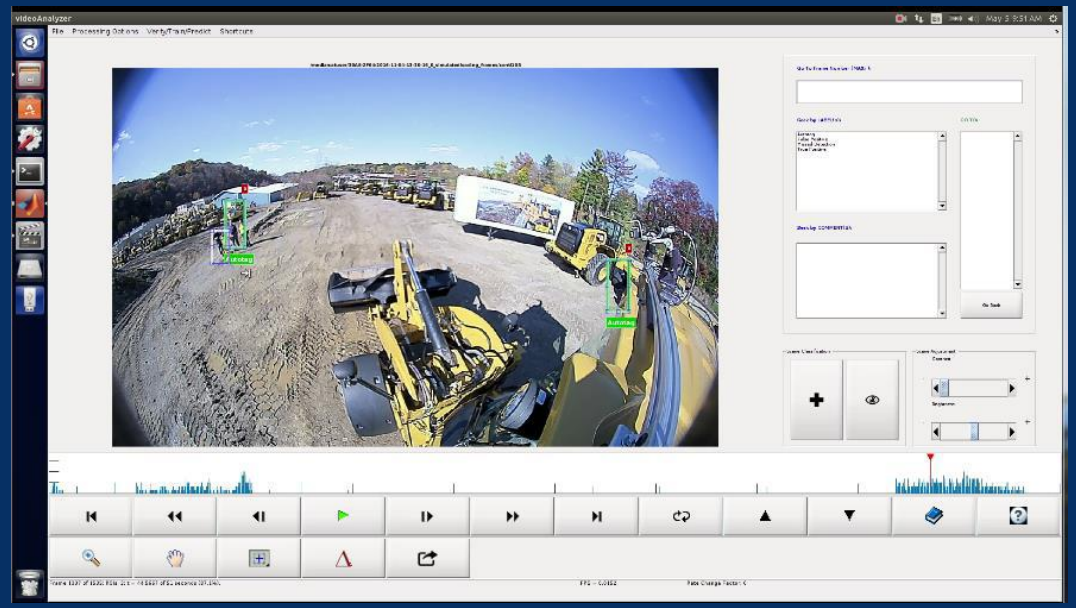
Dramatically reduce human supervision and development time

CATERPILLAR

- Partnered with MathWorks on their big data and machine/deep learning infrastructure
- Automatically ground-truths and labels data, reducing the need for human supervision and development time
- Tight integration with MATLAB for machine learning, visualization, and code generation

"We were spending way too much time hand-labeling our data…automatic detection and labeling of our data has been a tremendous advantage in efficiency."

— Larry Mianzo, Caterpillar



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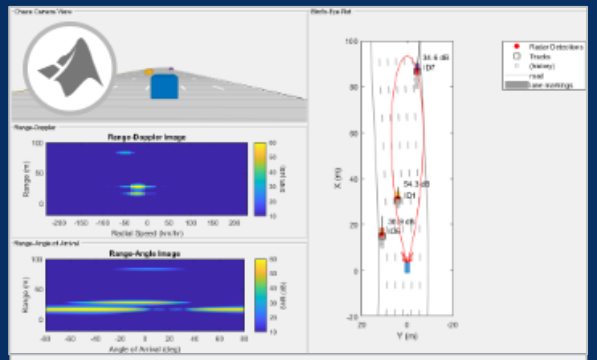
Generate synthetic data from Simulink to improve your datasets

Data Preparation

- Data cleansing and preparation
- Human insight
- Simulation-generated data

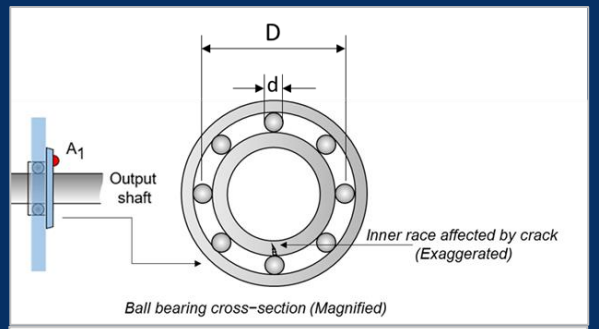


Radar Signal Simulation and Processing for Automated Driving

Automated Driving Toolbox
Phased Array System Toolbox

Visualize Automated Parking Valet Using 3D Simulation

Automated Driving Toolbox
Simulink



Simulate Ball Bearing Failure Using Simscape

Simulink
Simscape
Simscape Multibody

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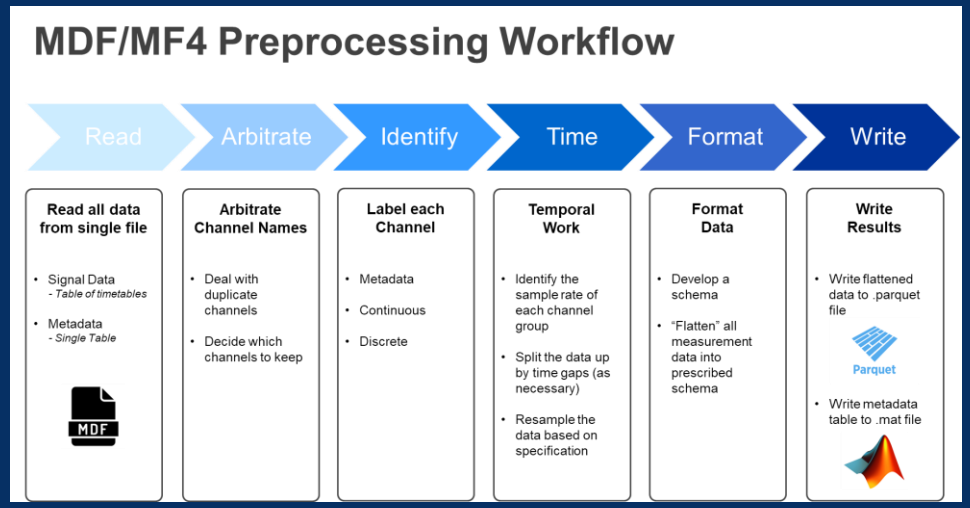
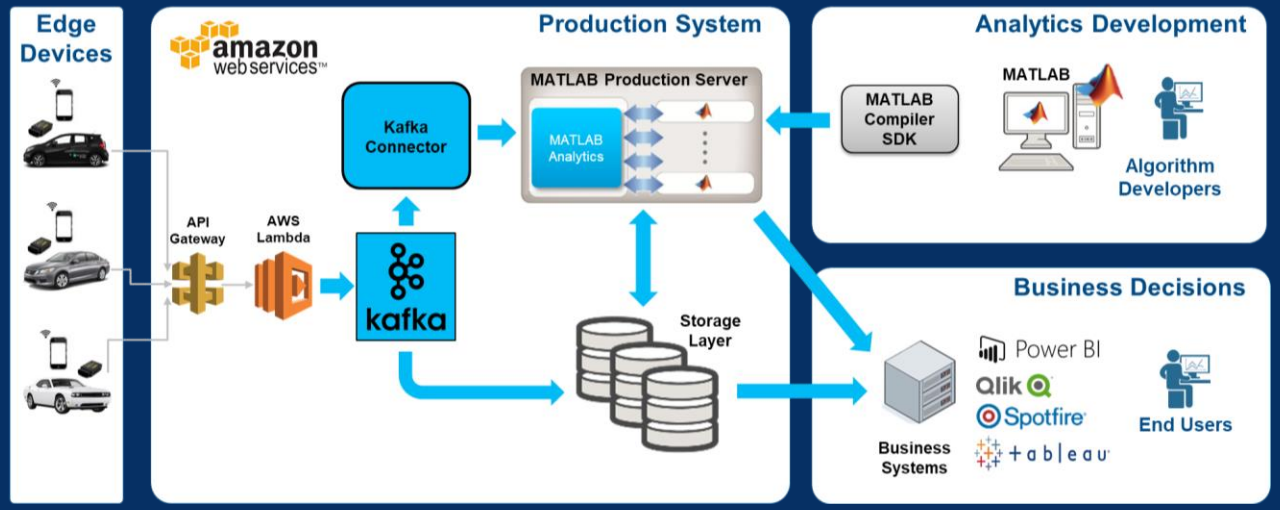
MathWorkers Help Customer Solving

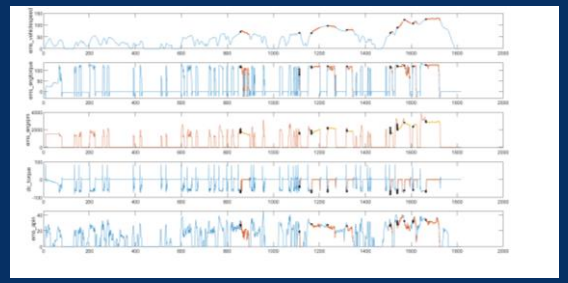
打通数据访问接口

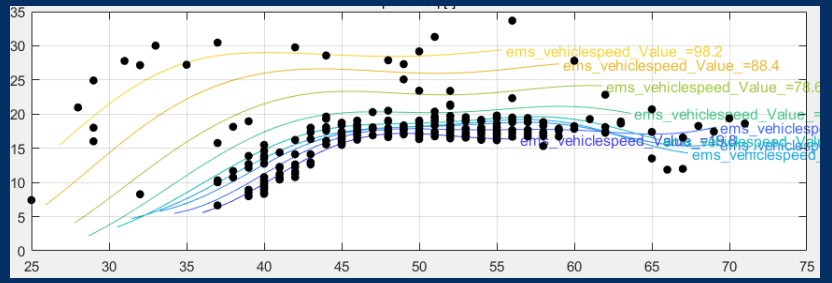


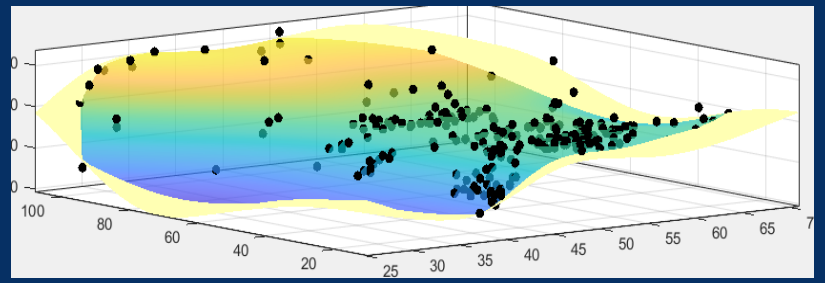
构建大数据开发平台



清洗数据并提取特征







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Start with a complete set of algorithms and pre-built models

Modeling

Model design and tuning

Hardware accelerated training

Interoperability

Algorithms

Machine learning
Trees, Naïve Bayes, SVM…

Deep learning
CNNs, GANs, LSTM, MIMO…

Reinforcement learning
DQN, A2C, DDPG…

Regression
Linear, nonlinear, trees…

Unsupervised learning
K-means, PCA, GMM…

Predictive maintenance
RUL models, condition indicators…

Bayesian optimization

Pre-built models

Image classification models
AlexNet, GoogLeNet, VGG, SqueezeNet, ShuffleNet, ResNet, DenseNet, Inception…

Reference examples

Object detection
Vehicles, pedestrians, faces…

Semantic segmentation
Roadway detection, land cover classification, tumor detection…

Signal and speech processing
Denoising, music genre recognition, keyword spotting, radar waveform classification…

…and more…

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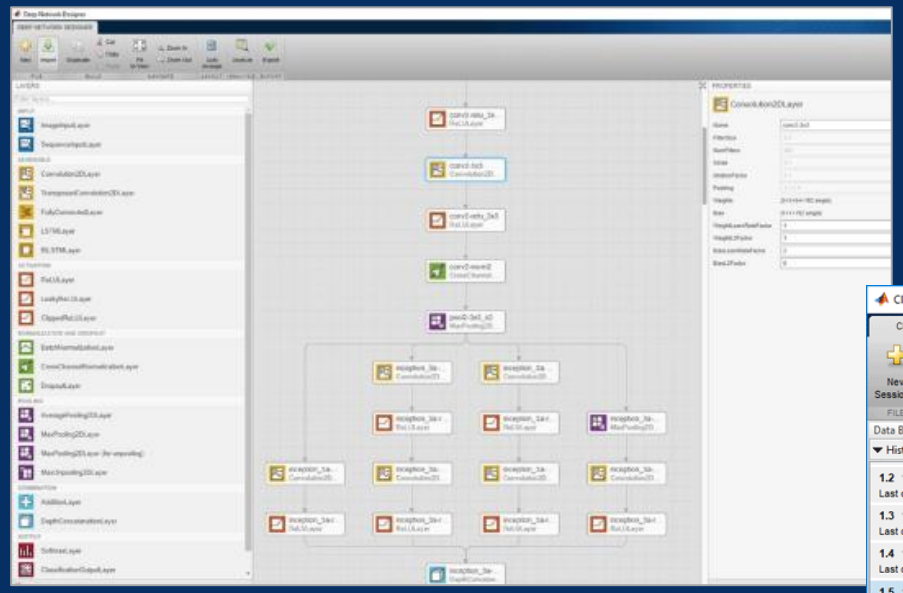
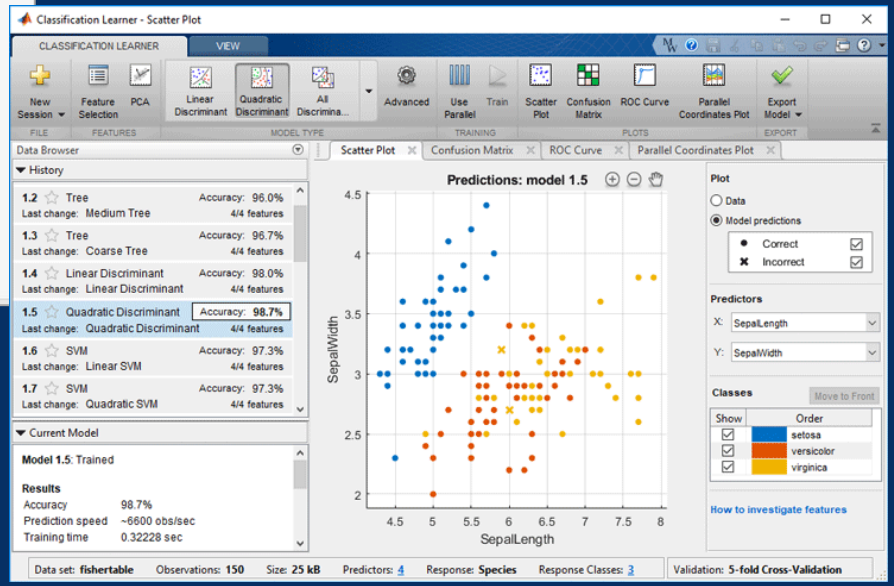
Apps for modeling turn your engineers into data scientists

Modeling

- Model design and tuning
- Hardware accelerated training
- Interoperability



Deep Network Designer app to build, visualize, and edit deep learning networks



Classification Learner app to try different classifiers and find the best fit for your data set

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Adapts to cluster, GPUs, cloud, and datacenter resources

Modeling

- Model design and tuning
- Hardware accelerated training
- Interoperability

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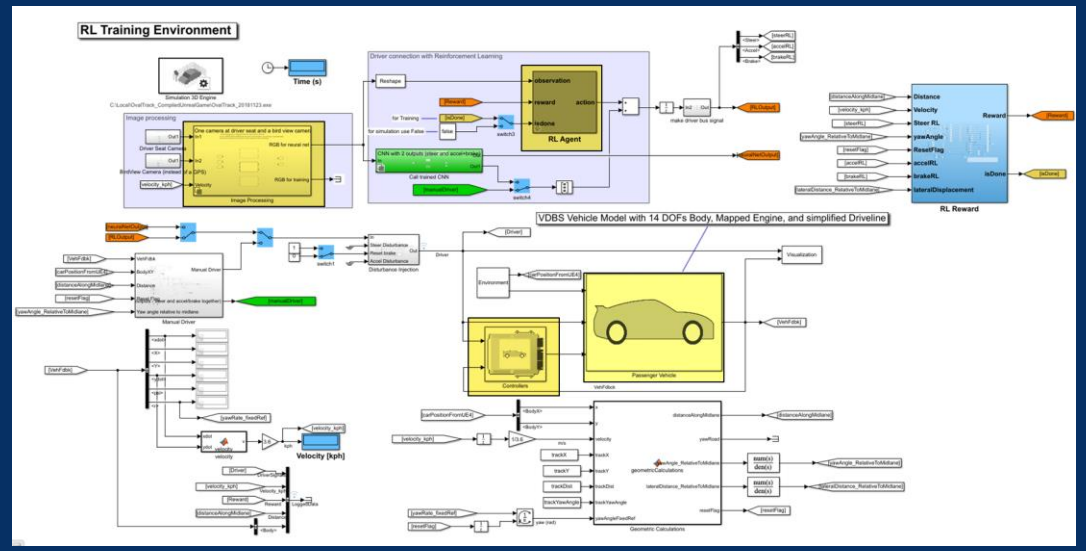
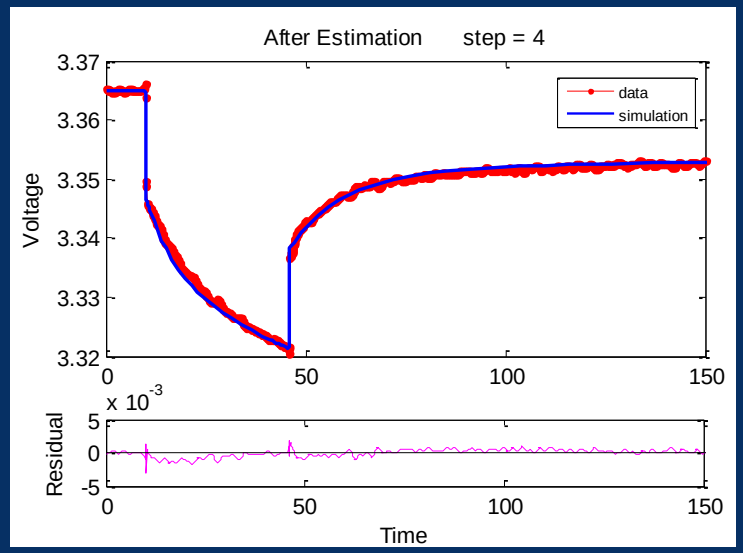
MathWorkers Help Customer Solving

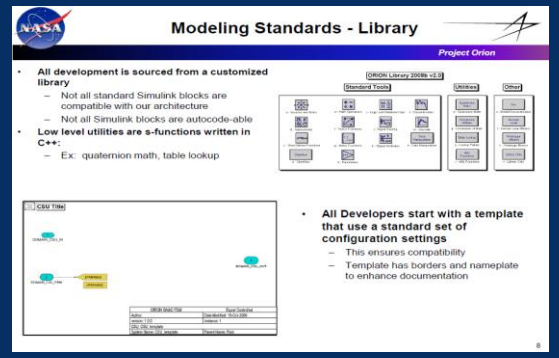
尝试新的方法

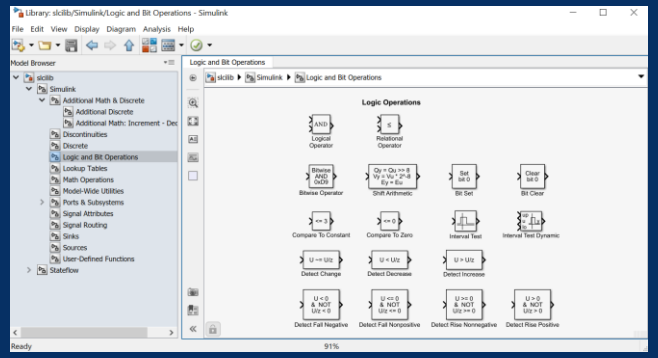


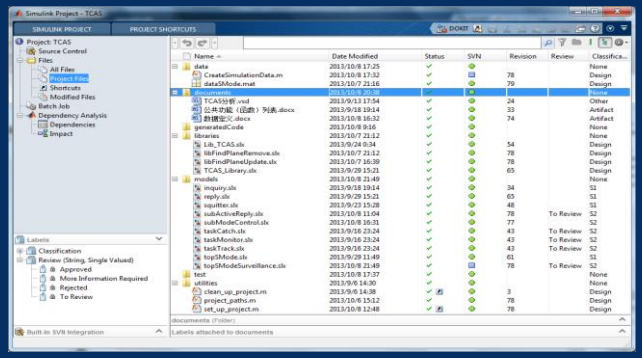
创建/精确化多域物理模型



优化模型架构、提高建模效率、促进持续集成……







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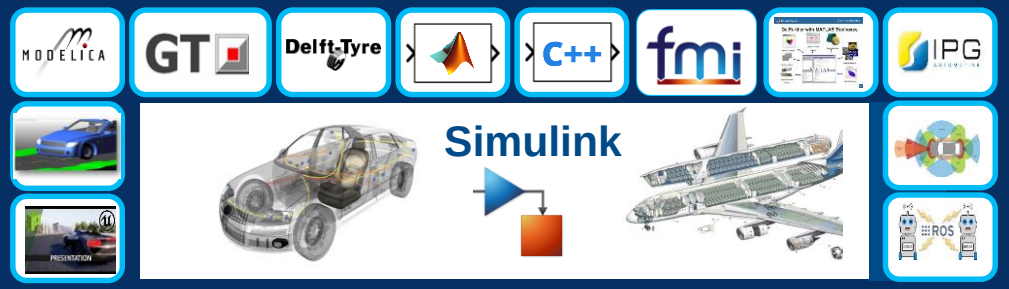
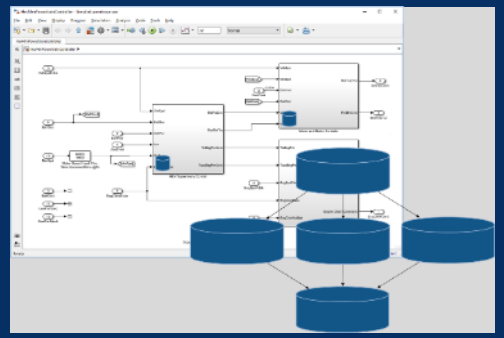
MATLAB as Simulation Platform Infrastructure

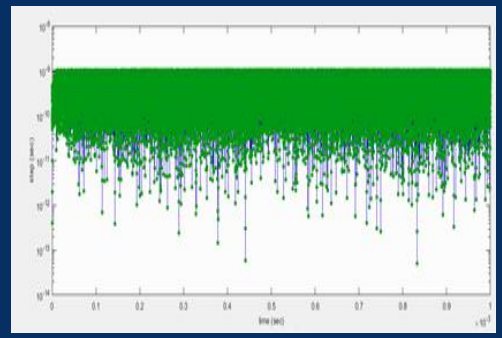


Simulation & Test

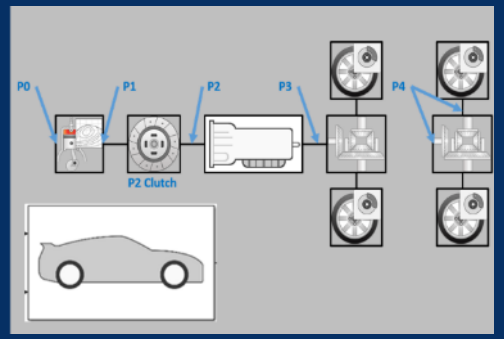
- Integration with complex systems
- System simulation
- System verification and validation



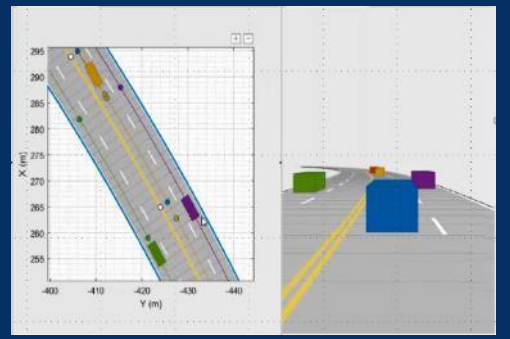
Data Management



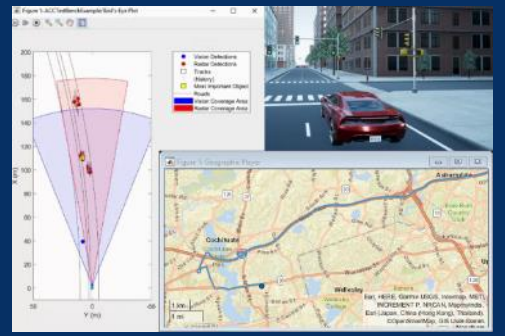
Solver Technology



Model Configuration



Scenarios



Visualization

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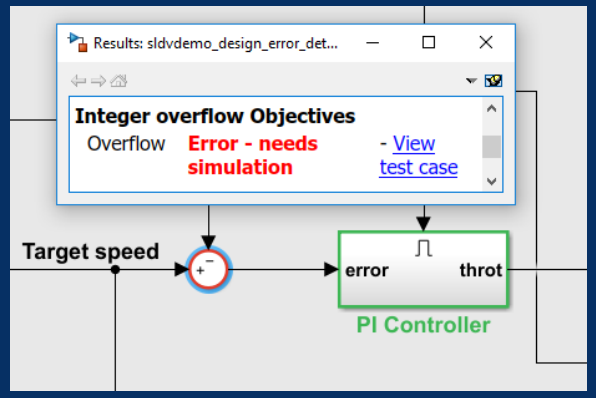
Automated Test and Verification

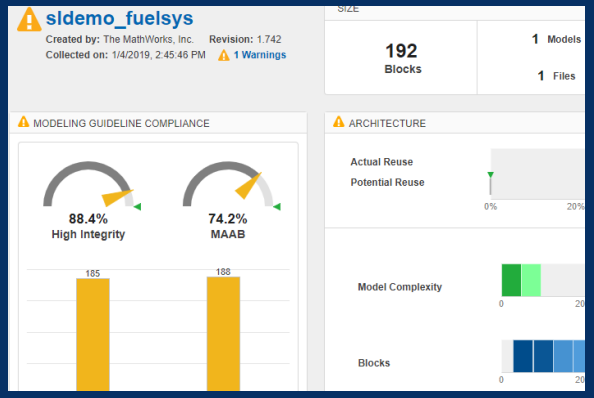
Simulation & Test

- Integration with complex systems
- System simulation
- System verification and validation



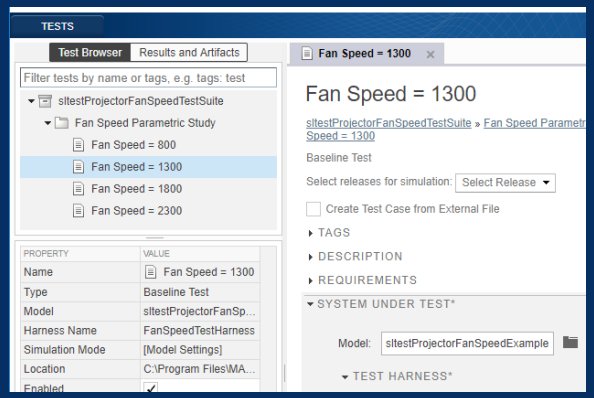
Find Bugs

Simulink Design Verifier
Polyspace Bug Finder



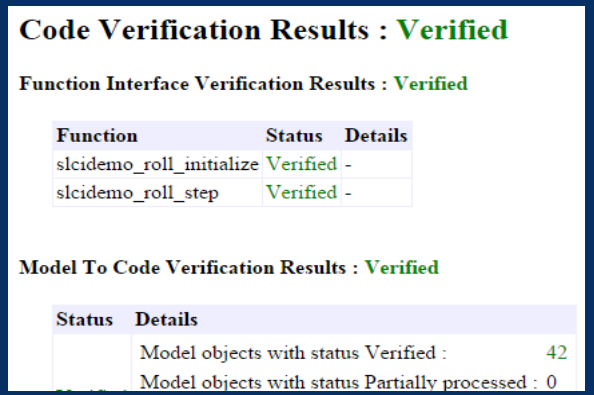
Check & Coverage

Simulink Check
Simulink Coverage



Manage Tests

Simulink Test



Inspect code

Simulink Code Inspector

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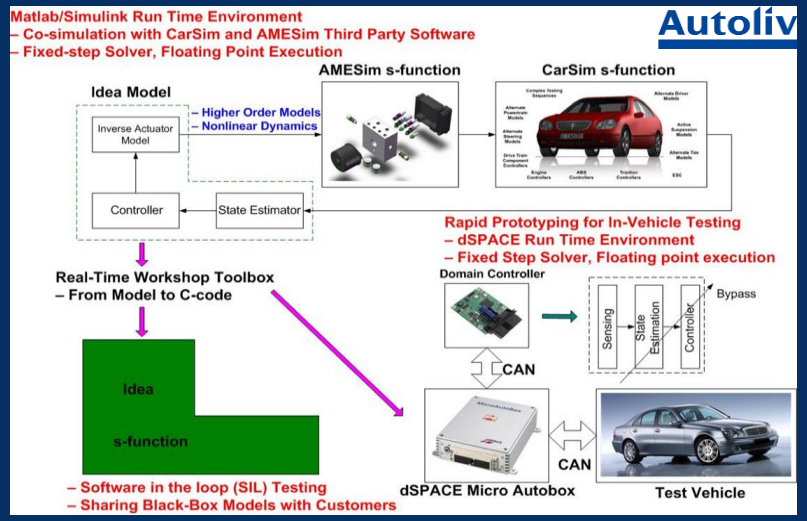
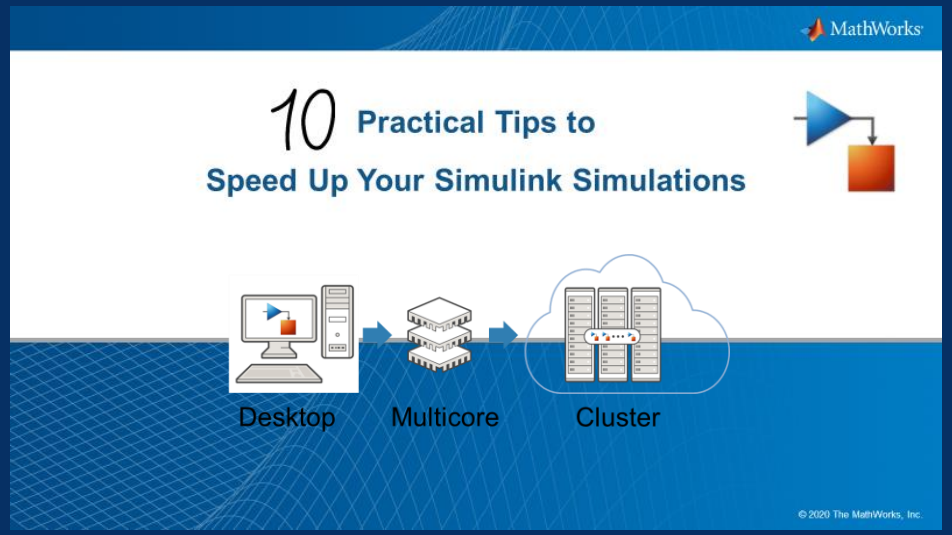
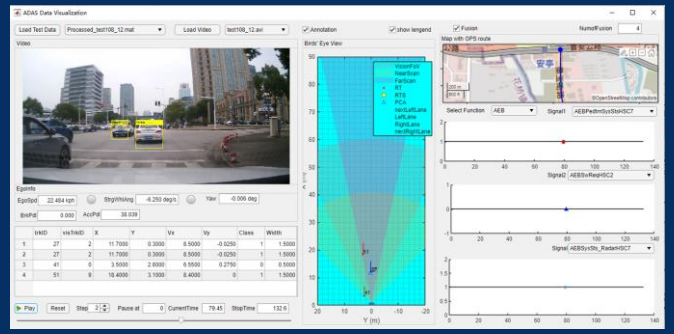
打通仿真接口/环节



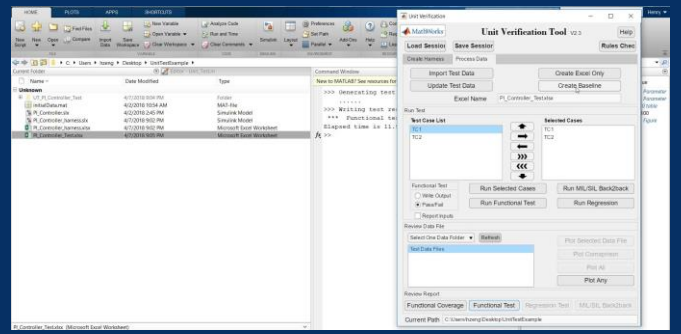
提高仿真效率



定制可视化App



定制自动化验证App



为行业认证提供帮助

Process Group		MathWorks Solution										
		Simulink	StateFlow	Embedded Simulink Coder	Simulink Requirements	System Composer	Simulink Test	Simulink Check	Simulink Design Verifier	Simulink Coverage	Polyspace Bug Finder	Polyspace Code Prover
System Engineering Process Group	System Requirements Analysis				○							
	System Architectural Design				○	○						
	System Integration/ Integration Test				○		○					
	System Qualification Test				○		○					
Software Engineering Process Group	Software Requirements Analysis				○							
	Software Architectural Design				○	○						
	Software Detailed Design	○	○				○	○	○			
	Unit Construction			○								
	Software Unit Verification						○			○	○	○
	Software Integration and Integration Test				○		○					
	Software Qualification Test				○		○					

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Deep experience in safety critical certification enables us to drive new standards for AI

Today

Tomorrow

EUROCAE WG-114 “Artificial Intelligence”

SAE G-34 “Artificial Intelligence in Aviation”

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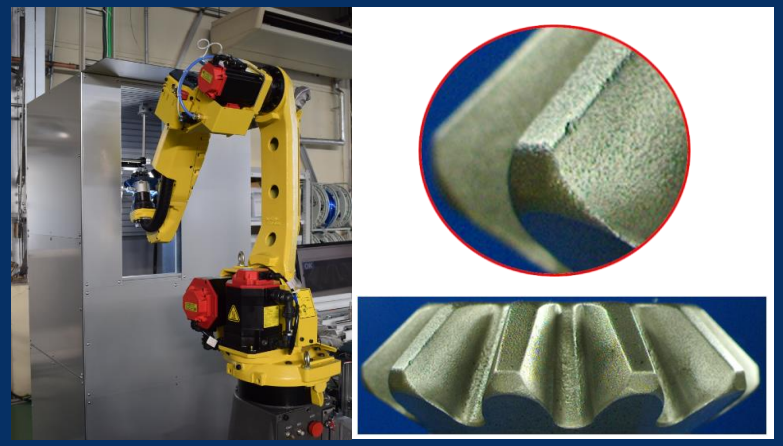
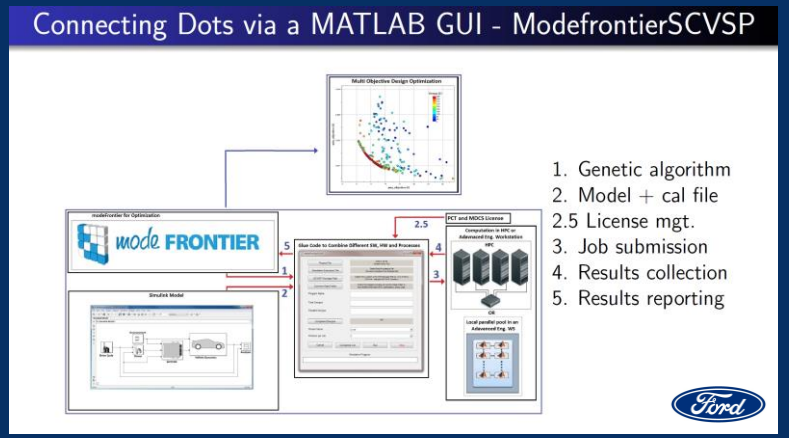
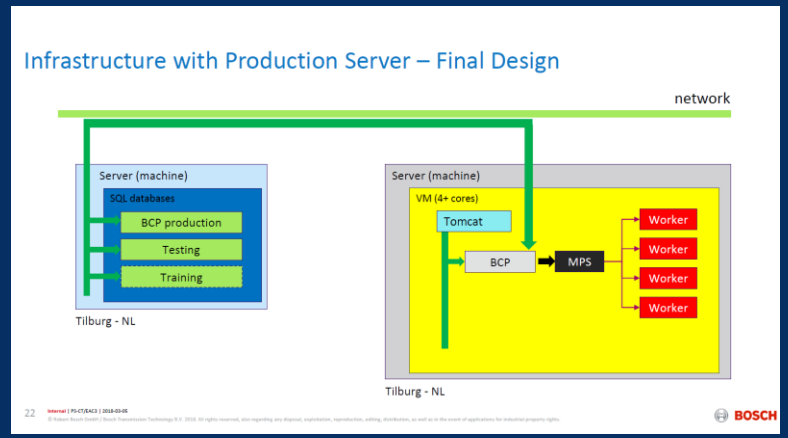
Models are useful everywhere

Deployment

- Embedded devices
- Enterprise systems
- Edge, cloud, desktop







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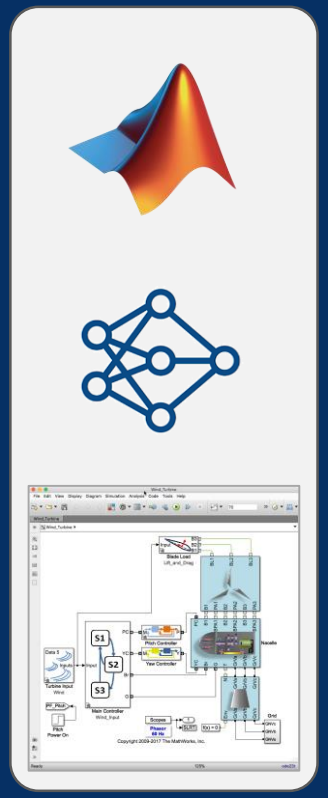
Deploy to any processor with zero coding errors

Deployment

- Embedded devices
- Enterprise systems
- Edge, cloud, desktop

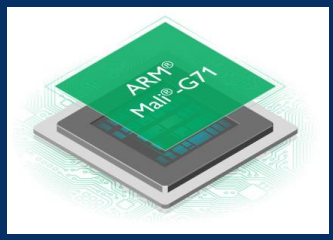


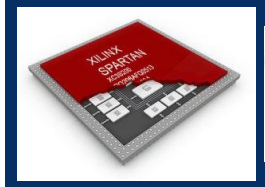
Code Generation

CPU

GPU

FPGA





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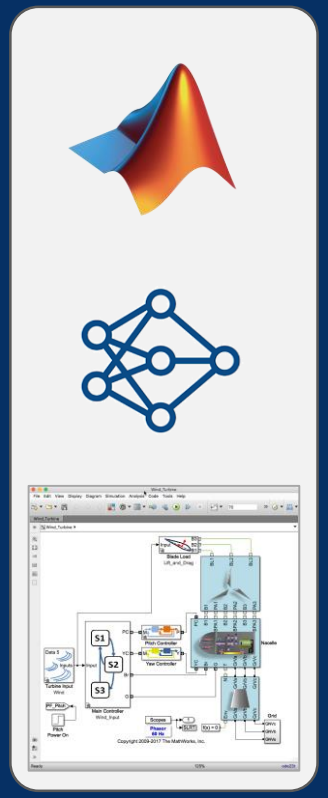
Deploy to enterprise IT infrastructure

Deployment

- Embedded devices
- Enterprise systems
- Edge, cloud, desktop



MATLAB Production Server

- Databases
- Streaming
- Cloud Storage
- Dashboards
- Containers
- Custom Tools

Cloud & Datacenter Infrastructure

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What's New: MATLAB Web App Server

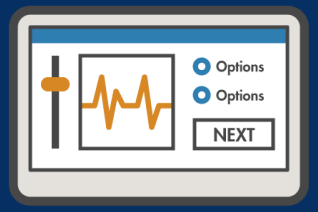
Deployment

- Embedded devices
- Enterprise systems
- Edge, cloud, desktop

Author

Domain Expert

Use App Designer to create MATLAB Apps



Use MATLAB Compiler to package

Package

Host

IT Administrator

Use MATLAB Web App Server to deploy and share

Access

End Users

Run web apps using a browser

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MathWorkers Help Customer Solving

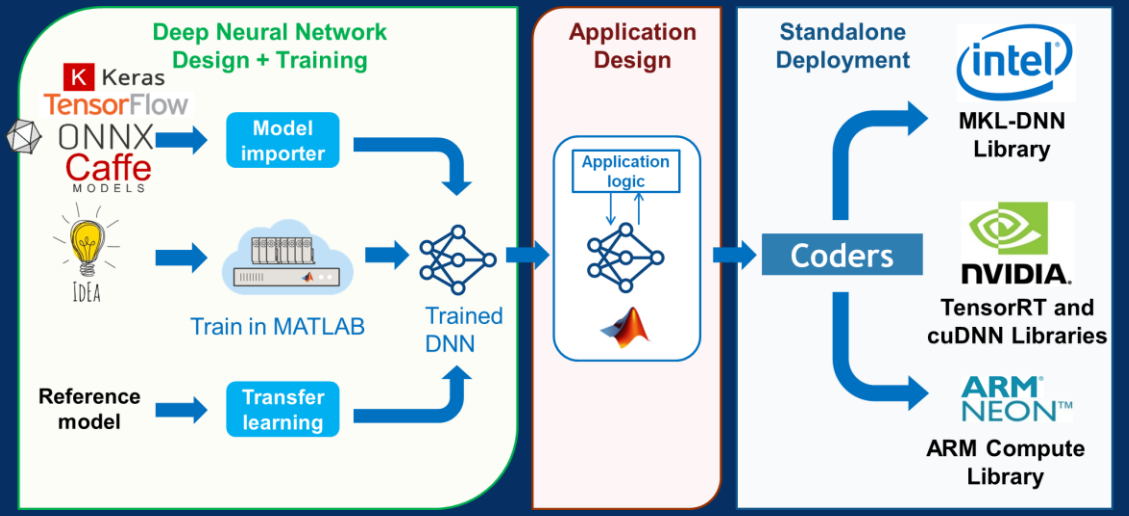
打通硬件接口

GPU
Fastest

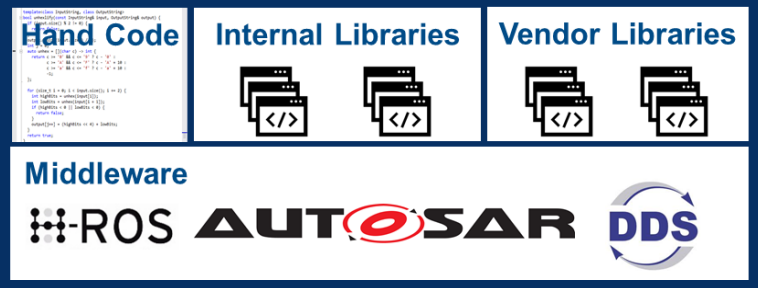
FPGA / ASIC
Lowest Power

MCU
Low Cost

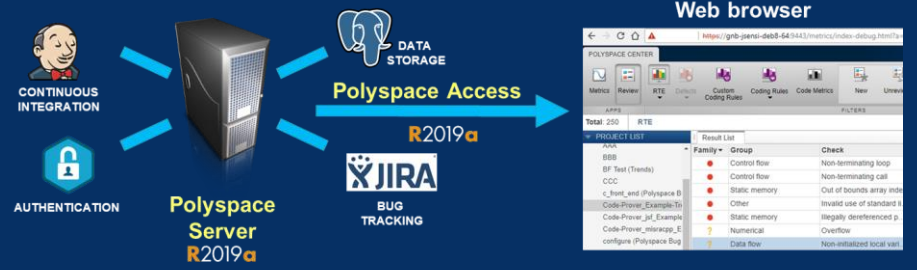
优化目标代码生成



为特定架构生成代码



开展持续的代码集成与验证



进行企业部署

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案例分享：创建精确的电池模型

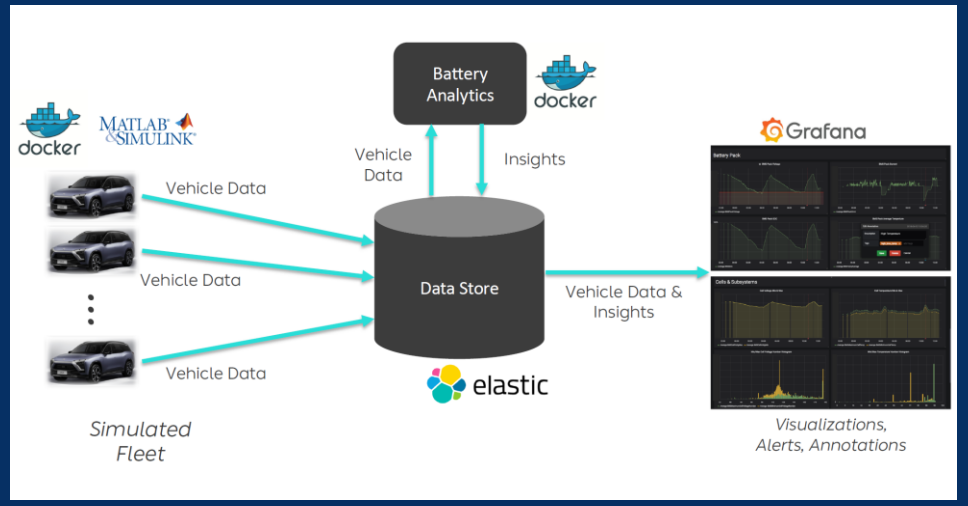
Tesla

Calibrated system model, improving simulation accuracy

Multidomain effects simulated, enabling dramatic advances in battery technology

https://www.mathworks.com/company/newsletters/articles/using-model-based-design-to-build-the-tesla-roadster.html

NIO



In the product design phase, battery data is available only under laboratory and limited driving conditions

Solution: Scalable simulation-based data generation deployed in the cloud

https://cn.mathworks.com/videos/building-battery-state-of-health-estimation-pipelines-for-electrified-vehicles-1558961478286.html

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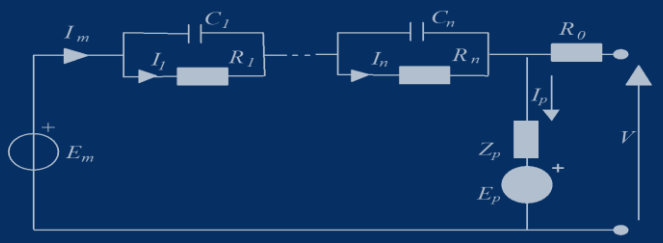
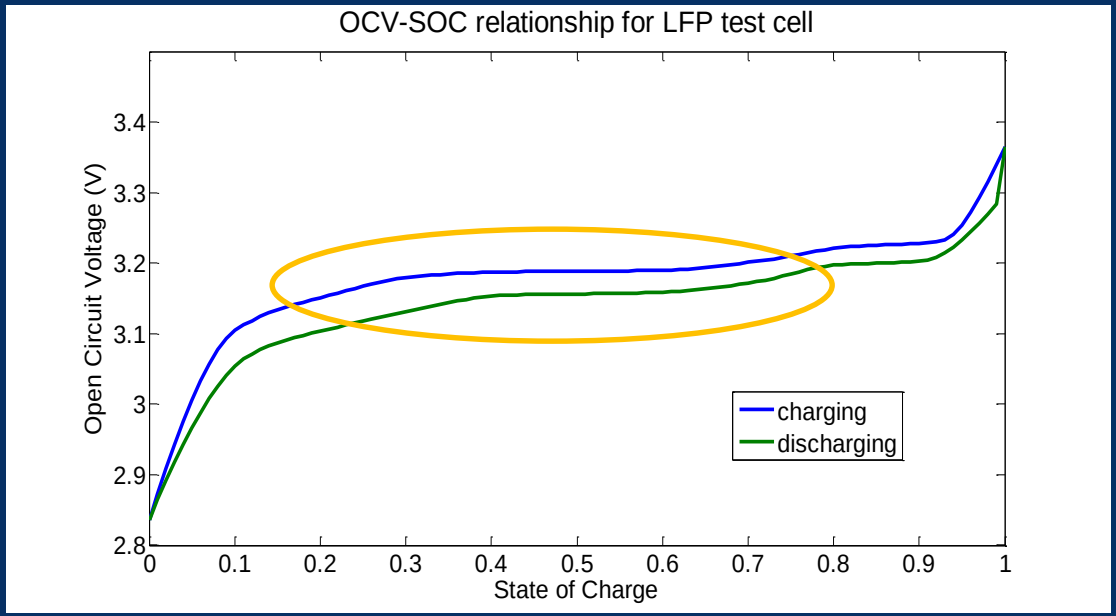
电芯建模存在的挑战

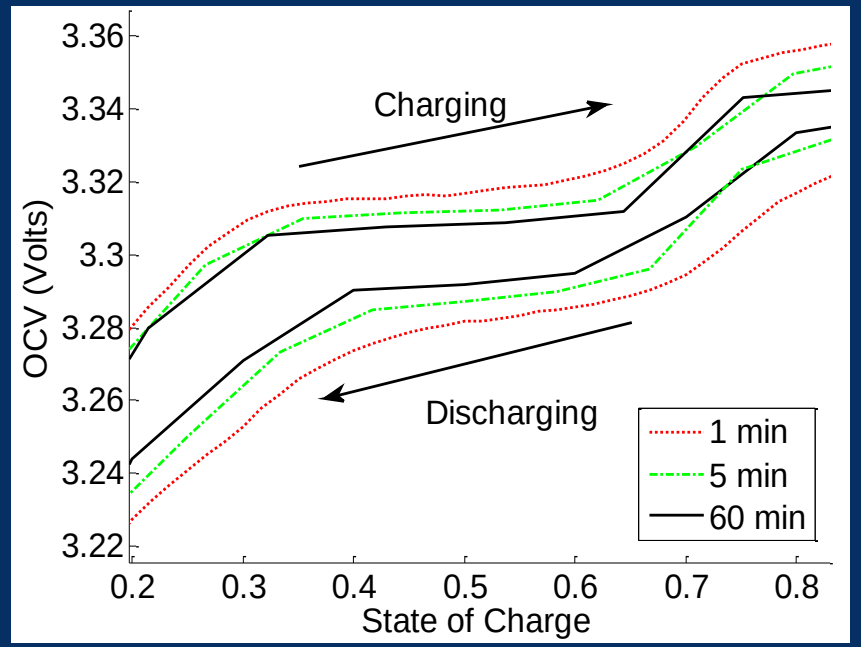
锂电池化学特性（以磷酸铁锂为例）：

- Complex transient response dynamics
- Significant voltage hysteresis
- Flat OCV-SOC relationship



$R_x = f(SOC, Temperature, I, V)$





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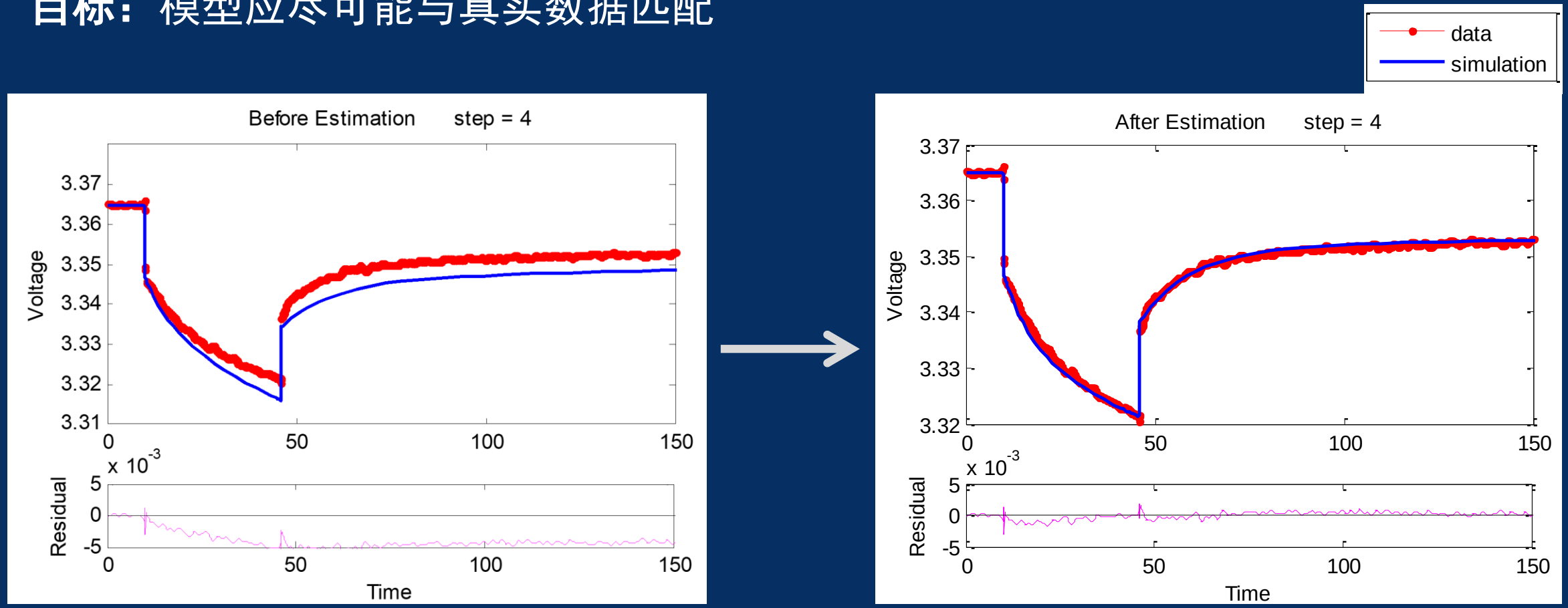
参数估计很复杂

目标：模型应尽可能与真实数据匹配



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精确化锂电池建模的工作流程

一. 设计HPPC脉冲实验

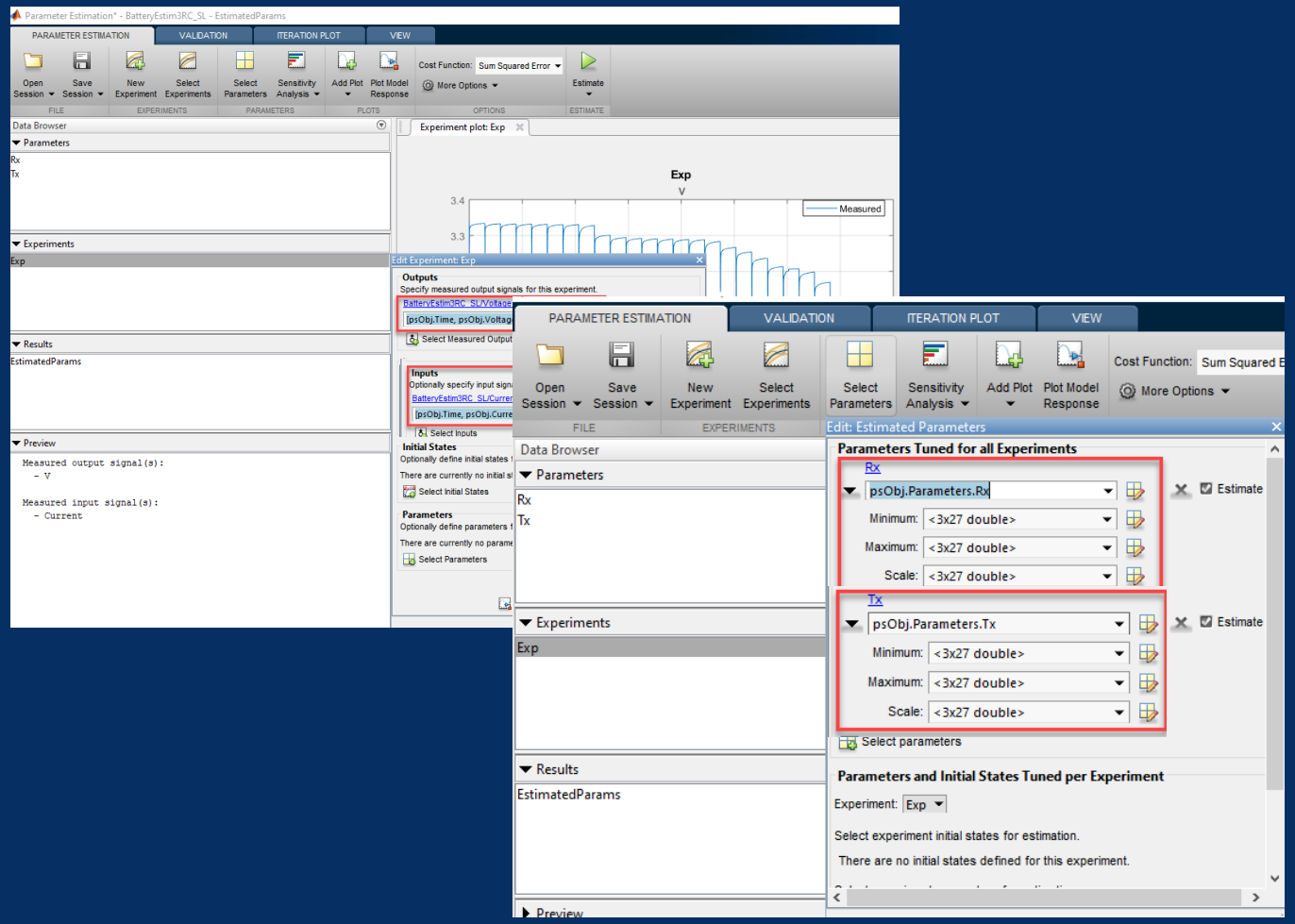
二. 实验数据预处理

三. 选取电路拓扑

四. 电芯参数估计

五. 数据平滑和后处理

六. 参数估计验证



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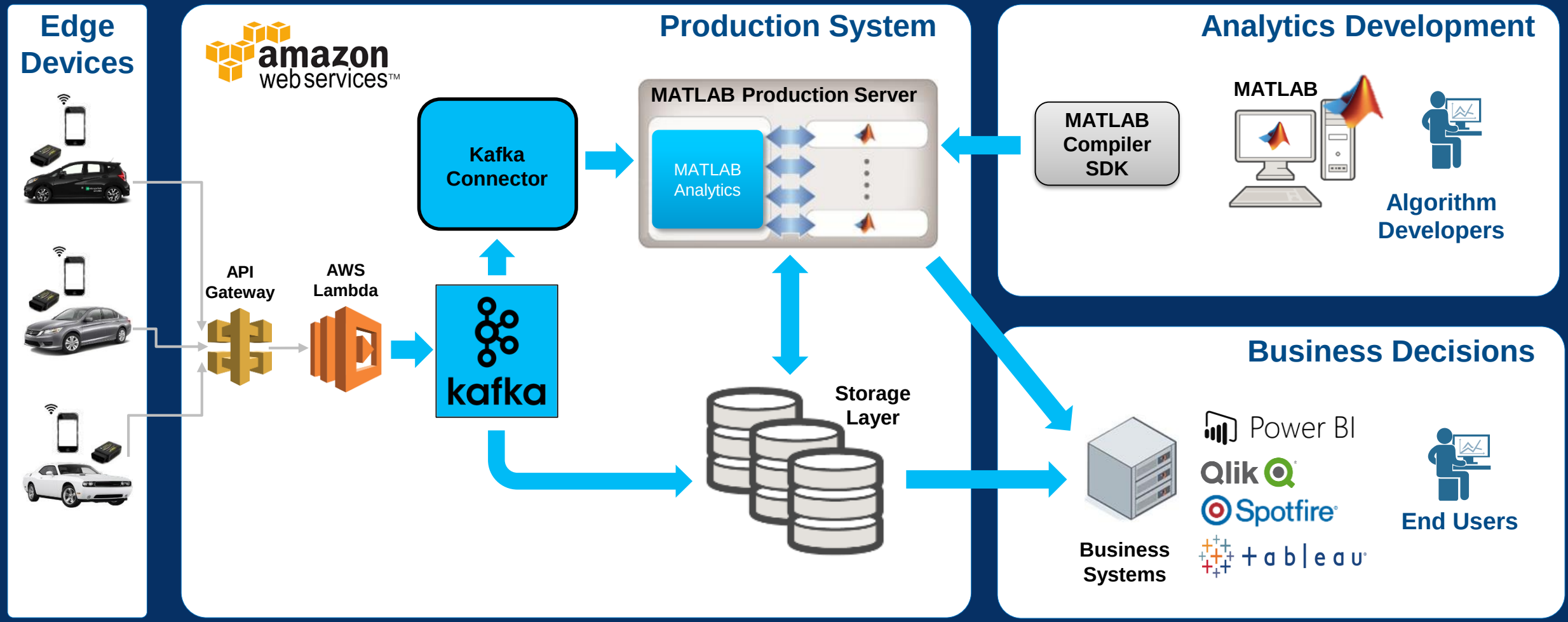
构建大数据与人工智能开发与应用平台



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Keep in mind today:

How can you systematically use models and data as part of your pragmatic digital transformation?

Partner with MathWorks !